Mathematical Analysis of a Deformable Prolate Half-Ellipse Model for Left-Ventricular Cross-Sectional Area Estimation

Abstract

This report presents a rigorous mathematical formulation and analysis of an interactive geometric boundary-tracing model for estimating the two-dimensional cross-sectional area of the left ventricle (LV) from three user-defined anatomical landmarks. The model represents the LV boundary as a deformable prolate half-ellipse (bullet-shaped) curve, constructed from an orthonormal local basis and parametrized trigonometrically, with local shape deformation introduced through shape-preserving piecewise cubic Hermite interpolation of user-specified control-point displacements. The enclosed area is evaluated numerically via the discrete Green's theorem (shoelace) formula and compared against the closed-form area of the underlying half-ellipse. Each stage of the pipeline is derived from first principles, a discretization error bound is established, computational complexity is analyzed, and degenerate and singular cases are characterized.

1 Problem Formulation

Two-dimensional echocardiographic estimation of left-ventricular cavity area from three landmarks — the apex and the two ends of the mitral annulus — is a discretized instance of the *area–length method*, in which the ventricular cross-section in an apical view is approximated by a half-ellipse (a "bullet" shape) [1]. The model analyzed here augments this rigid half-ellipse with a local, boundary-conforming deformation field, allowing the traced curve to depart from a pure ellipse while preserving smoothness. The problem decomposes into four mathematical sub-problems:

(i) construction of a local orthonormal coordinate frame from three landmark points;

(ii) parametric generation of a half-ellipse boundary in that frame;

(iii) smooth, shape-preserving deformation of the boundary from discrete control-point displacements; and

(iv) numerical evaluation of the area enclosed by the resulting closed curve.

2 Orthonormal Basis Construction

Let $P_1, P_2, P_3 \in \mathbb{R}^2$ denote the apex, base-left, and base-right landmarks respectively, expressed in image (pixel) coordinates in which the vertical axis increases downward. Define the base midpoint

$$M = \frac{P_2 + P_3}{2}. \tag{1}$$

The long-axis vector, its length (semi-major axis a), and the semi-minor axis b are

$$\vec{L} = P_1 - M, \qquad a = \|\vec{L}\| = \sqrt{(P_{1x} - M_x)^2 + (P_{1y} - M_y)^2}, \qquad b = \frac{\|P_3 - P_2\|}{2}. \tag{2}$$

The major-axis unit vector is $\hat{u} = \vec{L}/a$. An orthogonal candidate vector is obtained by a 90° rotation of $\hat{u}$,

$$\vec{v}_{\text{raw}} = R_{90^\circ}\,\hat{u} = (-u_y,\ u_x), \tag{3}$$

which is one of the two unit vectors orthogonal to $\hat{u}$. Since the sign of a 90° rotation is ambiguous with respect to which side of the long axis it points to, the correct orientation (toward P_2) is selected by a dot-product test:

$$\hat{v} = \begin{cases} \vec{v}_{\text{raw}}, & (P_2 - M)\cdot\vec{v}_{\text{raw}} \geq 0, \\ -\vec{v}_{\text{raw}}, & (P_2 - M)\cdot\vec{v}_{\text{raw}} < 0. \end{cases} \tag{4}$$

By construction, $\{\hat{u}, \hat{v}\}$ is an orthonormal basis ($\hat{u}\cdot\hat{v} = 0$, $\|\hat{u}\| = \|\hat{v}\| = 1$), and $(M, \hat{u}, \hat{v})$ defines a local Cartesian frame in which the half-ellipse is most naturally expressed.

Worked Example

Let $P_1 = (100, 10)$, $P_2 = (50, 110)$, $P_3 = (150, 110)$ (pixel units). Then

$$\begin{aligned} &M = (100, 110), && \vec{L} = P_1 - M = (0, -100), && a = 100, \\ &b = \tfrac{1}{2}\|(100, 0)\| = 50, && \hat{u} = (0, -1), && \vec{v}_{\text{raw}} = (-(-1),\ 0) = (1, 0). \end{aligned}$$

Testing orientation: $(P_2 - M)\cdot\vec{v}_{\text{raw}} = (-50, 0)\cdot(1, 0) = -50 < 0$, so $\hat{v} = (-1, 0)$, correctly pointing toward P_2. The analytic area of the resulting half-ellipse is computed in Section 5.

3 Parametric Half-Ellipse Boundary

Let $t \in [0, 1]$ be a normalized arc-length-independent parameter, with $t = 0$ corresponding to the apex and $t = 1$ to the base. Two symmetric angular parametrizations are used for the left and right halves of the boundary:

$$\theta_L(t) = \frac{\pi}{2}\,t, \qquad \theta_R(t) = -\frac{\pi}{2}\,t. \tag{5}$$

For either branch, the boundary point at parameter t is

$$P(\theta) = M + a\cos\theta\ \hat{u} + b\sin\theta\ \hat{v}. \tag{6}$$

This is the standard vector-form parametrization of an ellipse: setting $\hat{u} = (1, 0)$, $\hat{v} = (0, 1)$, $M = \mathbf{0}$ recovers $x = a\cos\theta$, $y = b\sin\theta$, i.e., (6) is that canonical ellipse subjected to a rigid rotation (encoded by $\hat{u}, \hat{v}$) and a translation (by M), with independent scaling a, b along the two orthogonal axes. At the endpoints, $P(0) = M + a\hat{u} = P_1$ and $P(\pm\pi/2) = M \pm b\hat{v} = P_2$ or P_3, confirming the curve interpolates all three landmarks exactly before any deformation is applied.

Each branch is discretized at N uniformly spaced parameter values $t_i = i/(N-1)$, $i = 0, \ldots, N-1$, producing an ordered set of boundary points $\{P(\theta(t_i))\}$.

4 Shape-Preserving Deformation via Piecewise Cubic Hermite Interpolation

To allow the traced boundary to depart locally from a pure ellipse (e.g. to follow a regionally non-elliptical wall), each arc admits two interior control points at $t = \frac{1}{3}$ and $t = \frac{2}{3}$, which may be displaced by a user-specified offset. Let

$$T = \left\{0,\ \tfrac{1}{3},\ \tfrac{2}{3},\ 1\right\}, \qquad V = \left\{0,\ \Delta_{1/3},\ \Delta_{2/3},\ 0\right\}, \tag{7}$$

where each Δ is a two-component planar displacement, and the endpoint offsets are fixed at zero so the apex and base landmarks remain interpolated exactly. Each Cartesian component of $\Delta(t)$ is interpolated independently by a Piecewise Cubic Hermite Interpolating Polynomial (PCHIP).

4.1 Formal Construction (Fritsch–Carlson)

For nodes $T_0 < T_1 < T_2 < T_3$ with data V_0, V_1, V_2, V_3, define the secant slopes and interval widths

$$\delta_k = \frac{V_{k+1} - V_k}{T_{k+1} - T_k}, \qquad w_k = T_{k+1} - T_k, \qquad k = 0, 1, 2. \tag{8}$$

The derivative d_k assigned to each interior node T_k ($k = 1, 2$) follows the Fritsch–Carlson shape-preserving rule [2]:

$$d_k = \begin{cases} 0, & \operatorname{sgn}(\delta_{k-1}) \neq \operatorname{sgn}(\delta_k) \text{ or } \delta_{k-1} = 0 \text{ or } \delta_k = 0, \\ \dfrac{3(w_{k-1} + w_k)}{\dfrac{2w_{k-1} + w_k}{\delta_{k-1}} + \dfrac{w_{k-1} + 2w_k}{\delta_k}}, & \text{otherwise,} \end{cases} \tag{9}$$

i.e. a weighted harmonic mean of the adjacent secant slopes when they agree in sign, and zero at any local extremum of the data. Endpoint derivatives ($k = 0, 3$) are set by a one-sided shape-preserving extrapolation, consistent with the standard PCHIP boundary treatment. On each subinterval $[T_k, T_{k+1}]$, the interpolant is the unique cubic Hermite polynomial matching (V_k, d_k) and (V_{k+1}, d_{k+1}):

$$H_k(t) = V_k\, h_{00}(s) + w_k\, d_k\, h_{10}(s) + V_{k+1}\, h_{01}(s) + w_k\, d_{k+1}\, h_{11}(s), \qquad s = \frac{t - T_k}{w_k}, \tag{10}$$

with the standard cubic Hermite basis functions $h_{00}, h_{10}, h_{01}, h_{11}$.

4.2 Shape-Preservation Property

The Fritsch–Carlson derivative rule guarantees that $\Delta(t)$ is C^1 and *monotone on any subinterval where the underlying data is monotone*, and in particular does not overshoot beyond the range of the control-point values. This distinguishes PCHIP from a natural (or not-a-knot) cubic spline, which minimizes the curvature functional $\int [S''(t)]^2\, dt$ and is only C^2-continuous: enforcing C^2 continuity is a global constraint that can force the interpolant to oscillate away from a purely local, monotone trend in the data (a manifestation of Runge-type behavior for closely spaced control data). Since the physical quantity being interpolated is a boundary displacement field that should smoothly increase or decrease between an anchor point and a control point without artificial dips, the C^1, monotonicity-preserving PCHIP interpolant is the appropriate choice over a C^2 spline.

The deformed boundary point is then

$$C(t) = P\big(\theta(t)\big) + \Delta(t). \tag{11}$$

5 Discrete Area via Green's Theorem

The discretized left and right arcs are concatenated (the right arc traversed in reverse) into a single closed polygonal curve with vertices $(x_1, y_1), \ldots, (x_n, y_n)$, $(x_n, y_n) = (x_1, y_1)$. By Green's theorem applied to the position vector field, the enclosed area equals the boundary line integral

$$A = \oint_{\partial\Omega} x\, dy = -\oint_{\partial\Omega} y\, dx = \frac{1}{2} \oint_{\partial\Omega} (x\, dy - y\, dx)\,, \tag{12}$$

which, evaluated on a piecewise-linear (polygonal) boundary, reduces to the discrete shoelace formula

$$A_{\text{numeric}} = \frac{1}{2}\left|\sum_{i=1}^{n-1}\left(x_i y_{i+1} - x_{i+1} y_i\right)\right|. \tag{13}$$

For the undeformed half-ellipse, the exact area follows directly by integrating the parametrization of Section 3 (or from the standard ellipse area formula πab, halved):

$$A_{\text{analytic}} = \frac{1}{2}\pi ab. \tag{14}$$

For the worked example of Section 2 ($a = 100,\ b = 50$), $A_{\text{analytic}} = \frac{1}{2}\pi(100)(50) \approx 7853.98$ square pixels.

5.1 Discretization Error Bound

For a smooth convex arc of total subtended angle Θ discretized into N equal-angle chords, each chord under-approximates the corresponding circular sector by the area of its circular segment. For a local radius of curvature r and chord angle $\delta\theta = \Theta/N$, the segment area is

$$\frac{r^2}{2}\left(\delta\theta - \sin\delta\theta\right) \approx \frac{r^2\,\delta\theta^3}{12} \qquad (\delta\theta \ll 1), \tag{15}$$

by the Taylor expansion $\sin x = x - x^3/6 + O(x^5)$. Summing over N chords on each of the two arcs, and approximating the local radius of the ellipse by $r \approx \sqrt{ab}$, gives the total area error

$$\varepsilon_{\text{area}} \;\approx\; \frac{2\,a\,b\,\Theta^3}{12\,N^2} \;=\; \frac{a\,b\,\Theta^3}{6N^2}. \tag{16}$$

For $a = 100,\ b = 50,\ \Theta = \pi/2,\ N = 150$: $\varepsilon_{\text{area}} \approx \dfrac{(100)(50)(\pi/2)^3}{6(150)^2} \approx 0.14$ square pixels, i.e. a relative error of order 10^{-5} (approximately 0.002%) relative to $A_{\text{analytic}} \approx 7854$. This confirms that, for $N = 150$ points per arc, the numerical shoelace area converges to the analytic half-ellipse area to a precision several orders of magnitude finer than any physically meaningful pixel- or landmark-placement uncertainty.

6 Computational Complexity

Table 1 summarizes the arithmetic cost of one full boundary/area update (e.g., following a control-point displacement), with N the number of discretization points per arc.

Stage	Operation	Cost
Basis construction (Sec. 2)	vector subtraction, norm, normalization	$O(1)$
PCHIP coefficients (Sec. 4)	Fritsch–Carlson slopes at 4 nodes	$O(1)$
Arc evaluation (Sec. 3)	$\cos\theta_i, \sin\theta_i$, affine combination	$O(N)$
PCHIP evaluation (Sec. 4)	cubic Hermite evaluation per point	$O(N)$
Shoelace summation (Sec. 5)	one multiply–add per vertex pair	$O(N)$

Table 1: Arithmetic complexity per boundary update, in terms of the per-arc discretization count N.

The overall complexity is therefore $O(N)$, dominated by the three linear-in-N stages; the basis and spline-coefficient stages contribute only a small, fixed additive constant independent of N. Since N is fixed at a small value (typically on the order of 10^2) and does not grow with problem size, the update cost in practice behaves as a small constant, which is what permits interactive (real-time) redraw rates during continuous control-point manipulation.

7 Degenerate and Singular Cases

7.1 Collinear or Coincident Landmarks

If P_1 coincides with M (equivalently, the apex lies on the line through P_2, P_3), then $a = \|\vec{L}\| \to 0$, and the normalization $\hat{u} = \vec{L}/a$ is undefined. This is a removable singularity only in the sense that the limiting geometry (a degenerate half-ellipse of zero major axis) is well-defined, but the normalized-vector computation itself is not; any implementation must treat $a < \epsilon$ for some small $\epsilon > 0$ as an invalid configuration rather than propagating a division by a near-zero quantity.

7.2 Non-Simple (Self-Intersecting) Polygons

The shoelace formula (13) is well-defined for *any* closed polygon, including non-simple (self-intersecting) ones, but its geometric interpretation changes: for a self-intersecting curve, the signed sum in (13) computes the algebraic sum of the signed areas of the constituent simple loops (positive for counterclockwise loops, negative for clockwise loops), not the total area enclosed by the curve's trace. Consequently, if a control-point displacement is large enough that the deformed boundary crosses itself, A_{numeric} remains a well-defined real number but no longer represents a physically meaningful enclosed area. This is a property of the shoelace formula itself (a consequence of Green's theorem applied to a non-simple curve, where the winding number about interior points may exceed one), not a numerical artifact, and is independent of the discretization density N.

8 Summary

The model reduces to four composable mathematical operations: (i) construction of an orthonormal local frame from three points via vector subtraction, normalization, and a sign-resolved 90° rotation; (ii) a rigid-rotation-and-scale parametrization of a canonical half-ellipse; (iii) a C^1, monotonicity-preserving cubic Hermite (PCHIP) deformation field anchored at the two landmark endpoints and free at two interior control points; and (iv) evaluation of the enclosed area via the discrete form of Green's theorem. The discretization error of the area computation is shown to be $O(1/N^2)$ and numerically negligible at the point counts used in practice, so that the dominant sources of uncertainty in an applied setting are external to the geometric model itself (pixel-to-physical calibration and landmark-placement variability) rather than internal to the numerical scheme.

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